Physical Chemistry (II)

Lecture 24

기말고사 안내 및 복습

기말고사 안내

- 시간: 12월 20일(수) 오후 12:30~오후 2:30
- 장소: 화학관 315호(홀수 학번), 114호(짝수 학번)
- 계산기 불필요
- 신분증 지참

시험 범위

강의 자료 기준 17강~23강

시험 문제

총 여섯 문제

- 과제 문제 및 그 변형(세 문제)
- 강의 자료에 나오는 내용 및 응용(두 문제)
- 보너스 문제(한 문제)

Overview

Multi-electron systems

Orbital approximation

Accuracy (Lec. 17-18)

- Hartree-Fock method
- Post-HF methods

Concepts (Lec. 19-23)

- Multi-electron atoms
- Molecules

Orbital Approximation

Each electron's spatial part is represented

by an one-electron orbital $\psi(\vec{r})$

- Typically $\psi(\vec{r})$ contains variational parameters
- If $\psi(\vec{r})$ is a linear combination of some functions,

$$\psi(\vec{r}) = c_1\phi_1 + c_2\phi_2 + \cdots + c_K\phi_K$$

  - Variational method ⟶ secular equation (cf. Lec. 20, pp. 10-15)
  - Basic concept behind basis sets

Hartree-Fock Method

- The self-consistent field (SCF) method
- The atomic wavefunction = Slater determinant

For N quantum states $a, b, \cdots, n$,

$$\phi_{ab\cdots n}(1,2,\cdots,N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

Extension of HF

- Start from one-electron wavefunctions ("orbitals") u_i

$$\phi = |u_a \quad u_b \quad \cdots \quad u_n|$$

- HF determinants use occupied orbitals only

- Inclusion of unoccupied (virtual) orbitals may increase accuracy and take care of correlation
- Configuration interaction (CI), coupled-cluster (CC) theory

Multi-Electron Systems

H atom	1-electron wf ("orbitals")	H_2^+ molecule
He atom	2 electrons in orbitals	H_2 molecule
Li, Be, ...	*N* electrons in orbitals	He_2^+, He_2, ...

Not necessarily precise

Atoms and Molecules

	Atoms	Diatomic molecules	Polyatomic molecules	Conjugated systems
Construction of orbitals	L19, 12-13	L20	L22 L23, 10-17	L23, 19-25
Electron configurations	L19, 14	*Same Aufbau principle*		
Electronic states	L19, 15-28	L21, 14-25	L23, 18	
Applications	L19, 29-35	L21, 26-32		L23, 26-30

Atoms

- Orbitals: based on H-atom wavefunctions
- Electron configurations: Aufbau principle
- Electronic states (for light elements): term symbols
- Ground state: Hund's rules
- Applications
  - Selection rules → spectroscopy
  - Relativistic effects

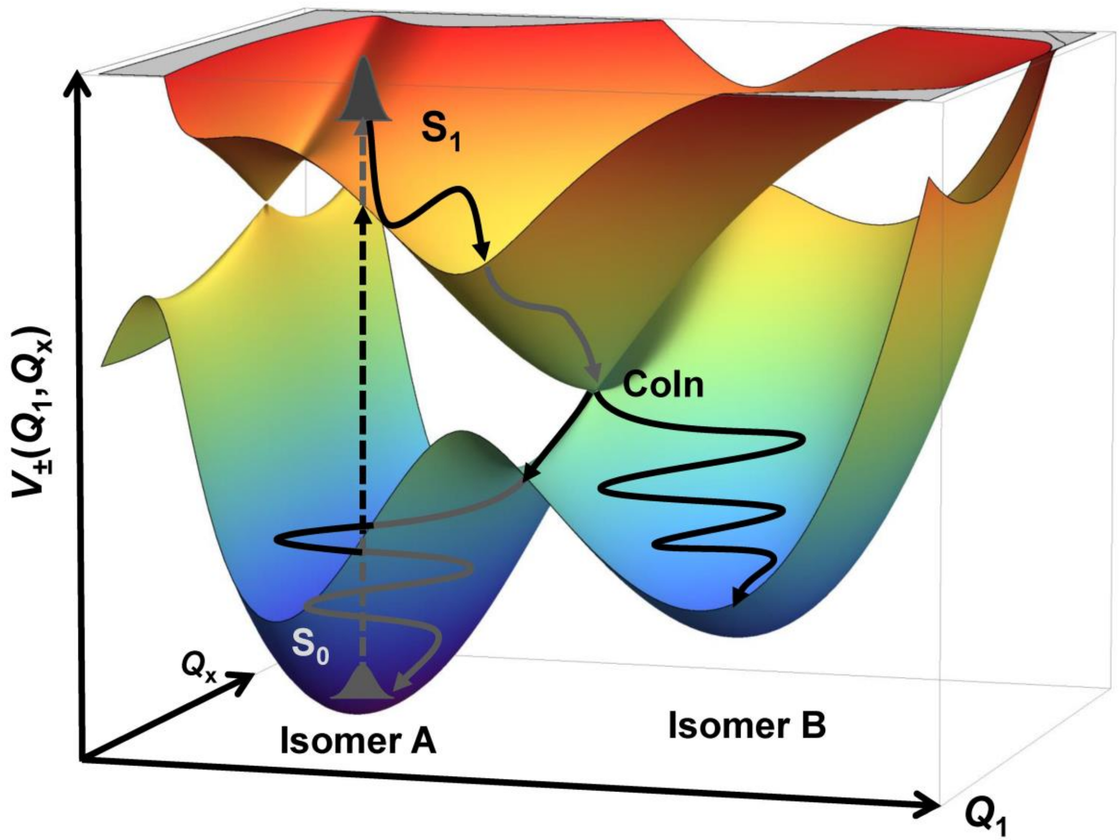
Diatomic Molecules

- Orbitals: based on atomic orbitals
  - LCAO-MO theory and VB theory
- Electron configurations: Aufbau principle
- Electronic states: term symbols
  - Potential energy curves
- Ground state: Hund's rule 1
- Applications
  - Molecular vibrations
  - Fate of excited states

(Small) Polyatomic Molecules

- Orbitals: based on atomic orbitals
  - LCAO-MO theory (group theory) and VB theory (hybridization)
- Electron configurations: Aufbau principle
- Electronic states
  - Potential energy surfaces

Potential Energy Surfaces

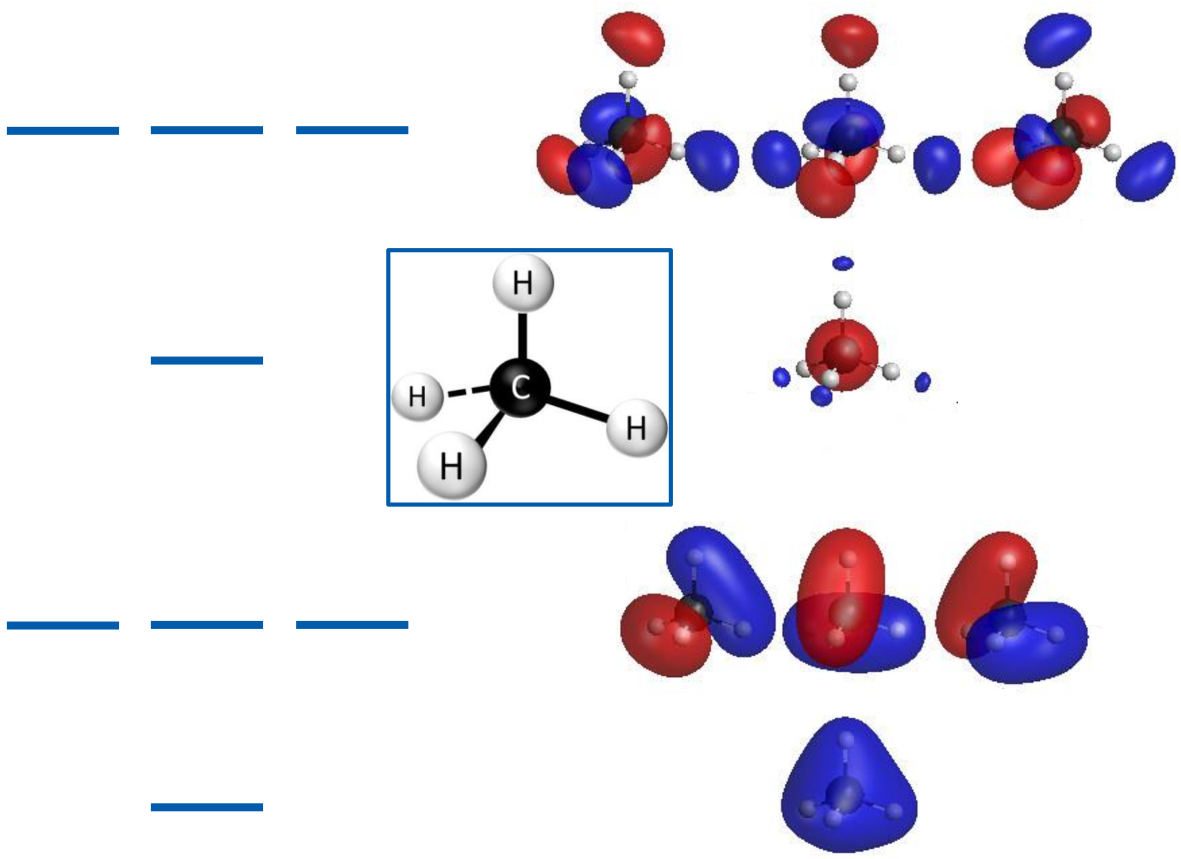


(Image: https://www.temps.phc.uni-kiel.de/en/research/laboratories-and-instrumentation/femtosecond-spectroscopy)

Conjugated Systems

- Orbitals: based on π orbitals
  - Hückel MO theory
- Electron configurations: Aufbau principle
- Applications
  - UV-visible spectroscopy
  - Aromaticity and antiaromaticity

MOs and Chemical Bonds



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